# Determinate Beam Calculation Report

IndeterminateBeam

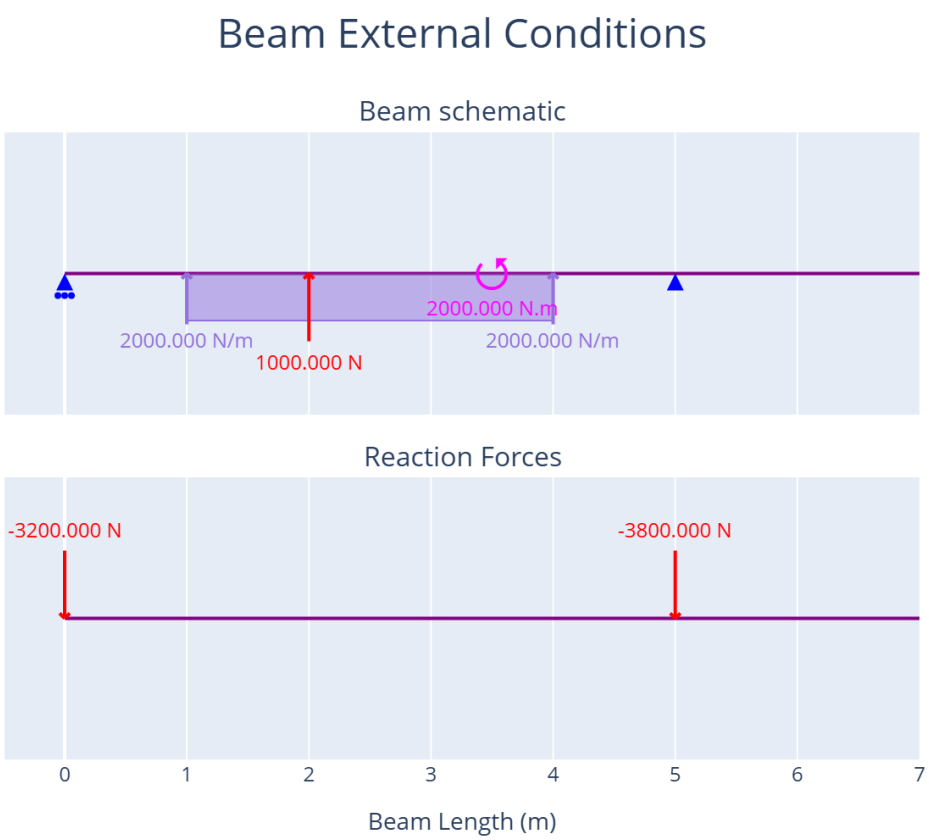


Figure 1: Beam schematic with applied loads and solved reaction forces.

## 1 Problem description

### 1.1 Beam

Span length $L = 7.0$ m.

### 1.2 Supports

- roller at $x = 0$ m

- pin at $x = 5$ m

### 1.3 Loads

- Point load $P = 1000$ N at $x = 2$ m
- Distributed load from $x = 1$ to $x = 4$ m (length 3 m)
- Point torque $M = 2000$ N.m at $x = 3.5$ m

## 2 Equilibrium equations

For a statically determinate beam, the three equations of equilibrium (in the plane) are sufficient to solve for the three reaction unknowns. Each equation is expanded explicitly below.

**Sum of forces in $x$:**

$$R_{x,5} \text{ (reaction at } x = 5 \text{ m)} \quad x_5$$

$$\sum F_x = 0 \quad \Rightarrow \quad x_5 = 0$$
$$\Rightarrow \quad x_5 = 0 \text{ N}$$

**Sum of forces in $y$:**

| | |
|---|---|
| Point load $P = 1000$ N at $x = 2$ m | $= 1000$ N |
| Distributed load total (integrated) 6000 N | $= 6000$ N |
| $R_{y,0}$ (reaction at $x = 0$ m) | $y_0$ |
| $R_{y,5}$ (reaction at $x = 5$ m) | $y_5$ |

$$\sum F_y = 0 \quad \Rightarrow \quad y_0 + y_5 + 7000.0 = 0$$

**Sum of moments (about $x = 0$):**

| | |
|---|---|
| Point load $P \times x = 1000 \times 2$ N.m | $= 2000$ N.m |
| Distributed load moment about 0 15000 N.m | $= 15000$ N.m |
| Point torque $M = 2000$ N.m | $= 2000$ N.m |
| $R_{y,0} \times 0$ N.m | $= 0$ N.m |
| $R_{y,5} \times 5$ N.m | $5y_5$ |

$$\sum M_0 = 0 \quad \Rightarrow \quad 5y_5 + 19000.0 = 0$$
$$\Rightarrow \quad y_5 = -3800 \text{ N}$$

## 3 Solution

Solving the linear system yields the following reactions:

| Support ($x$) | $R_x$ | $R_y$ | $M$ |
|---|---|---|---|
| 0 m | 0 N | $-3200$ N | 0 N.m |
| 5 m | 0 N | $-3800$ N | 0 N.m |

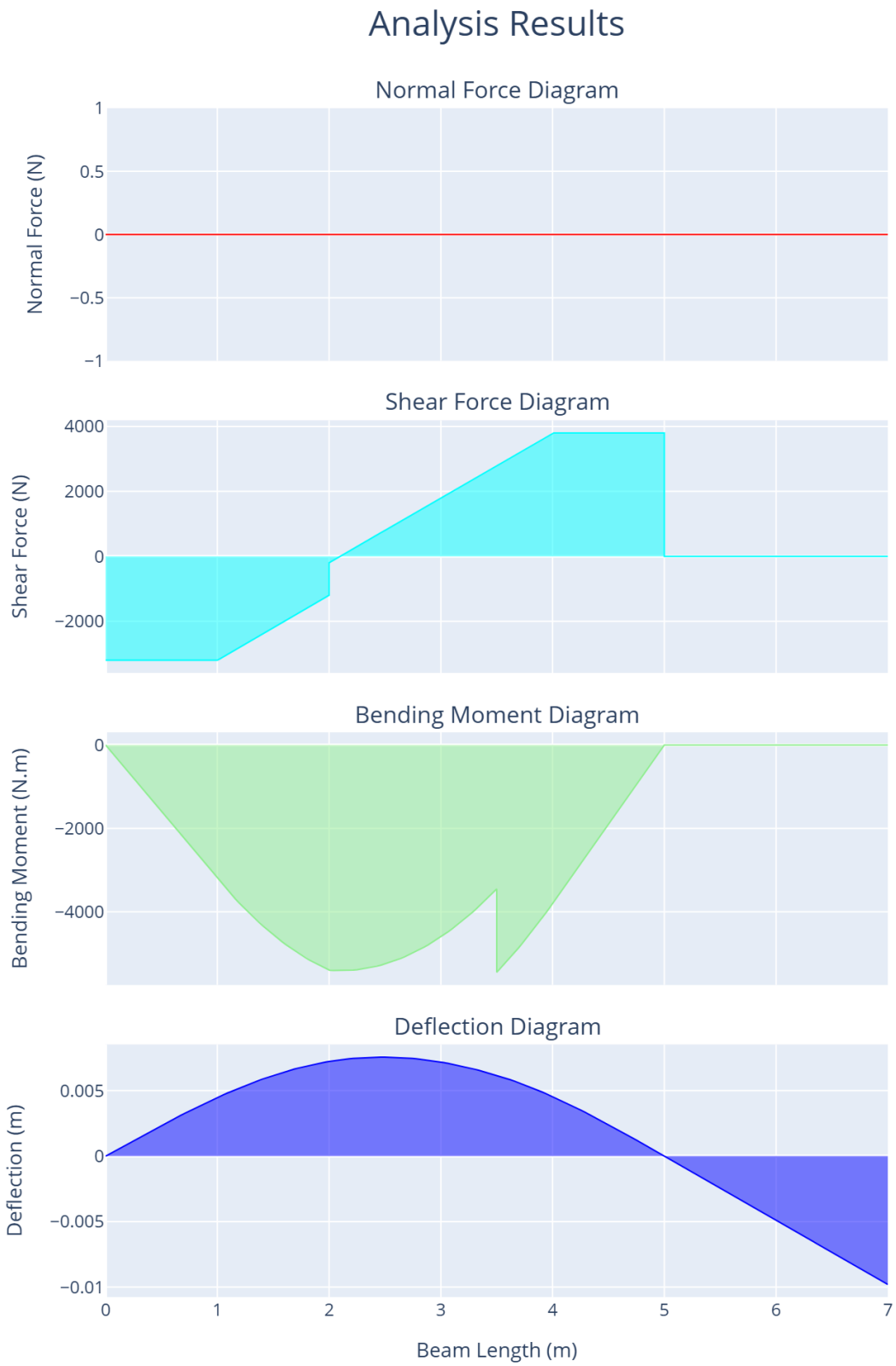


Figure 2: Internal forces and deflection: normal force, shear force, bending moment, and deflection diagrams.